

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Analyzed structured and unstructured data from semiconductor manufacturing processes to identify root causes of yield deviations, enabling targeted corrective actions and improving overall process health by **15%**.
- Collaborated with Technology Development and Local Yield teams to facilitate the import of new semiconductor technologies into production fabs, ensuring successful product qualification and adherence to yield targets.
- Managed product maintenance lifecycles, overseeing the introduction and optimization of products within fabrication lines to meet key targets for yield, device performance, and operational efficiency.
- Developed and implemented workflows and procedures for continuous improvement programs, reducing yield excursions in line and sort processes by **12%** through enhanced quality control.
- Assessed manufacturing materials and process changes for quality and reliability risks, ensuring strict adherence to Intel's internal standards and regulatory requirements.
- Extracted actionable insights from complex datasets using **Python** (pandas, NumPy) and advanced statistical analysis, supporting data-driven decision-making for process improvement activities.

SKILLS

Data Analysis & Machine Learning – Data Mining, Statistical Analysis, Root Cause Analysis, Yield Analysis, Python (pandas, NumPy, scikit-learn), R, Regression, Hypothesis Testing, Time-Series Analysis, Machine Learning Fundamentals

Semiconductor Manufacturing – FEOL/BEOL Integration, Device Engineering, Defect Reduction, Lithography, Advanced Patterning, Dry Etch, Wet Cleaning, Diffusion, Thin-Film Deposition, CMP, Process Engineering, Quality Assurance, Reliability

Software Engineering & Cloud – Java Spring Boot, Python, REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Shell Scripting, Git, Jenkins, CI/CD

Data Visualization & BI – Power BI, DAX, Power Query, Tableau, Data Modelling, Data Warehousing, BI Dashboards, Operational Reporting

Collaboration & Communication – Cross-functional Teams, Stakeholder Collaboration, Technical Presentations, Written Communication, Teamwork, Agile/Scrum, Production Support, Incident Response

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Yield Optimization Dashboard for Semiconductor Operations

- Developed a comprehensive dashboard using **Power BI** to visualize yield performance metrics and defect trends across critical semiconductor manufacturing stages.
- Integrated data from multiple sources including fab process logs and defect inspection systems, creating robust **data models** and DAX measures for real-time analysis.
- Automated data refresh processes via **Power Query**, enabling operations teams to monitor process health and proactively identify excursions, reducing manual reporting effort by **40%**.

Automated Data Pipeline for Process Health Monitoring

- Engineered a **Python**-based ETL pipeline to ingest and transform large volumes of manufacturing data, preparing it for analysis and visualization.
- Implemented data validation checks and error handling within the pipeline, ensuring data integrity for downstream analytic models and reporting.
- Deployed the pipeline on **AWS Lambda**, integrating with **S3** for data storage and scheduled execution via CloudWatch Events, improving data availability for analysis by **99.9%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)